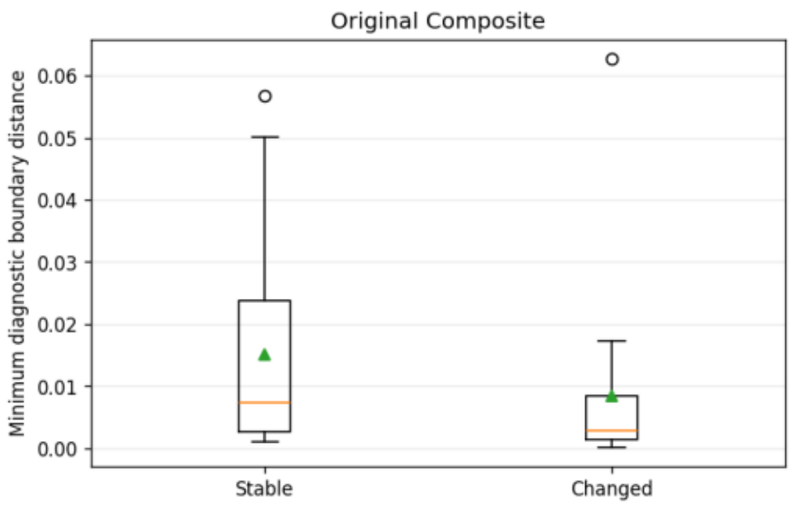
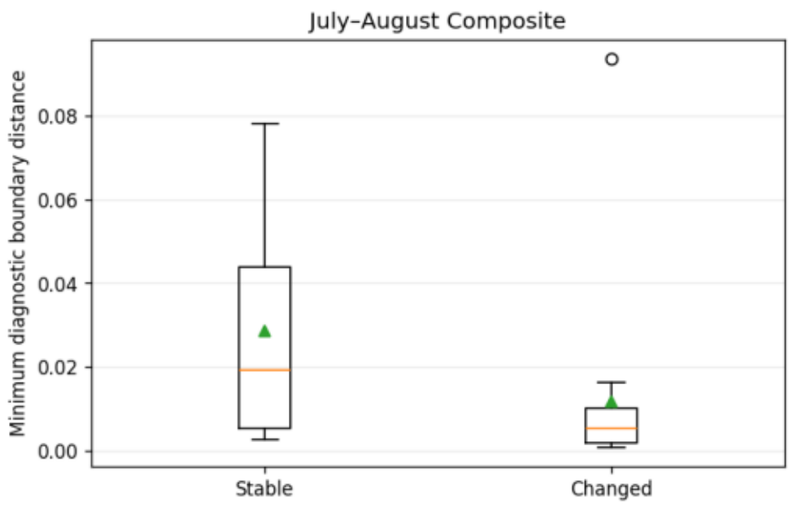
AGRICULTURAL SUSTAINABILITY ASSESSMENT

Ate Makoce | Environmental Data Cube for Oglala Lakota

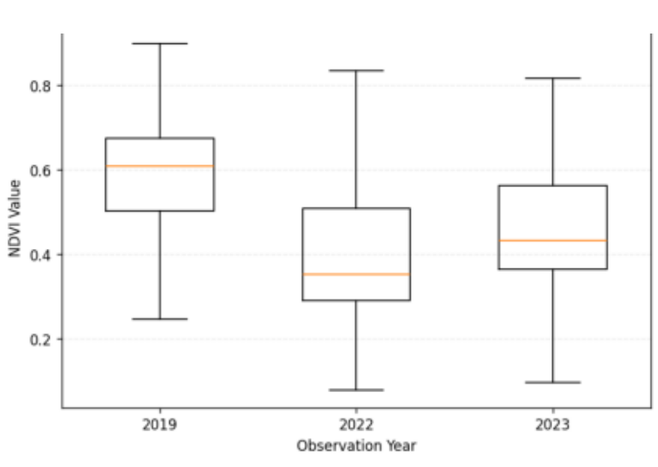
A | Original Composite



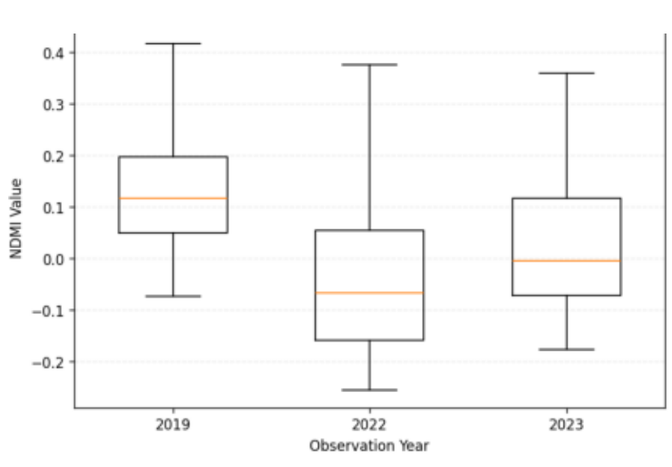
B | July–August Composite



C | NDVI Distribution



D | NDMI Distribution



RESULTS INTERPRETATION

A–B | Classification stability	Stable candidates generally show greater diagnostic boundary distances and broader variability than Changed candidates. The July–August composite shows a wider distribution among Stable candidates. These comparisons describe boundary proximity, not classification accuracy.
C | Vegetation greenness (NDVI)	Median NDVI is highest in 2019, declines in 2022, and partially recovers in 2023, indicating variation in observed vegetation greenness.
D | Moisture-related conditions (NDMI)	Median NDMI is positive in 2019, lower in 2022, and closer to zero in 2023, indicating variation in moisture-related spectral conditions.

Interpretive limitation: Satellite indices and diagnostic boundary distances do not independently establish overgrazing, degradation, or management causation.

Source: Notebook 09 | SA 16.17.5E and SA 18.2 | Original saved results; analyses not rerun.